through the ribosome by three bases. This expels deacylated / uncharged tRNA from P-site and moves peptidyl tRNA into the P-site and empties the A-site. The mRNA and tRNA are moved through the ribosome is called **translocation**. The translocation requires the hydrolysis of GTP.

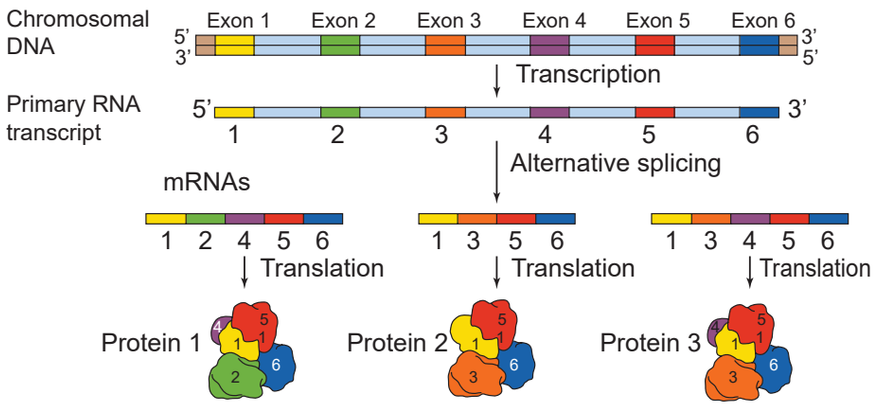


Figure: 3.24 Alternative splicing in plants

The ribosome (ribozyme - peptidyl transferase) catalyses the formation of peptide bond by adding amino acid to the growing polypeptide chain.

The ribosome moves from codon to codon along the mRNA in the 5' to 3' direction. Amino acids are added one by one translated into polypeptide as dictated by the mRNA. Translation is an energy intensive process. A cluster of ribosomes are linked together by a molecule of mRNA and forming the site of protein synthesis is called as **polysomes** or **polyribosomes.**

3. Termination of polypeptide synthesis

Eukaryotes have cytosolic proteins called **release factors** which recognize the termination codon, UAA, UAG, or UGA when it is in the A site. When the ribosome reaches a stop codon the protein synthesis comes to an end. So ribosomes are the protein making factories of a cell. When the polypeptide is completed the ribosome releases the polypeptide and detaches from the mRNA molecule. Now the ribosome splits into small and large subunits after the release of mRNA.

3.7.4 Alternative Splicing in plants

It is very useful in regulating gene expression to overcome the environmental stress in plants.

Alternative splicing is an important mechanism / process by which multiple mRNA's and multiple proteins products can be generated from a single gene. The different proteins generated are called isoforms. There are various modes of alternative splicing. When multiple introns are present in a gene, they are removed separately or as a unit. In certain cases one or more exons which is present between the introns are also removed.

Significance of alternative splicing

1. The proteins transcribed from alternatively spliced mRNA containing different amino acid sequence lead to the generation of protein diversity and biological functions.
2. Multiple protein isoforms are formed.
3. It creates multiple mRNA transcripts from a single gene. A process of producing related proteins from a single gene thereby the number of gene products are increased.
4. It plays an important role in plant functions such as stress response and trait selection. The plant adapts or regulates itself to the changing environment.

3.7.5. RNA Editing – Post Transcriptional RNA Processing in plants

Transcriptional RNA Processing in plants

Chemical modification such as base modification, nucleotide insertion or deletions and nucleotide replacements of

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